

Exercise #10

20. March 2023

Problem 1.

Compute the Laplace transform of the following functions.

a) $f(t) = 1 + 2t - 4t^4$.

b) $f(t) = 2t^2 e^{3t}$.

c) $f(t) = e^{-2t} \sin(5t)$.

Problem 2.

Find the inverse Laplace transform of the following functions.

a) $F(s) = \frac{3}{s} + \frac{1}{s^8}$.

b) $F(s) = \frac{2s + 1}{s^2 - s - 2}$

c) $F(s) = \frac{1}{(s + 1)(s^2 + 1)}$.

Problem 3.

Decide for each of the following statements whether it is true or false. Explain your answer.

- a) If f and g are two functions for which the Laplace transform exists, then $\mathcal{L}(f + g) = \mathcal{L}(f) + \mathcal{L}(g)$.
- b) If f and g are two functions, for which the Laplace transform exists, then $\mathcal{L}(f \cdot g) = \mathcal{L}(f) \cdot \mathcal{L}(g)$.
- c) If the function f satisfies $0 \leq f(t)$ for all $t \geq 0$, then $\mathcal{L}(f)(s) \geq 0$ for all s for which $\mathcal{L}(f)(s)$ exists.
- d) If the function f is continuous and satisfies $0 \leq f(t) \leq 1$ for all $t \geq 0$, then $\mathcal{L}(f)(s)$ exists for all $s > 0$.
- e) If the function f satisfies $0 \leq f(t) \leq 1$ for all $t \geq 0$, then $\mathcal{L}(f)(s)$ exists for all $s > 0$.

Problem 4.

Use the Laplace transform in order to solve the following initial value problems:

- a) $2y'' + y' - y = 0$, $y(0) = 0$, $y'(0) = 1$.
- b) $y'' + y = 1$, $y(0) = 0$, $y'(0) = 0$.
- c) $2y'' + 10y' + 12y = 0$, $y(0) = -2$, $y'(0) = 1$.

The next exercises are optional and should not be handed in!

Problem 5.

Compute the Laplace transform of the following functions:

- a) $f(t) = \sin(4t)e^{3t}$.
- b) $f(t) = \sinh(t) \cos(t)$.

Problem 6.

Compute the inverse Laplace transform of $F(s)$, where

a) $F(s) = \frac{1}{s^2 + s + 1},$

b) $F(s) = \frac{as}{s^2 - 2as + a^2 + 1}$ for some $a \in \mathbb{R},$

c) $F(s) = \frac{1}{s^3 + s}.$

Problem 7.

a) Use the Laplace transform to solve the differential equation,

$$y' = -2y + 1, \quad y(0) = 0.$$

b) Use the Laplace transform to solve the differential equation,

$$y''' = -y' + 2, \quad y(0) = y'(0) = y''(0) = 0.$$